25901338 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCHPY200 - APPLIED PHYSIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Abnormal increase in blood volume is called: [U] / [CO1]

   a). Hypovolemia b). Hypervolemia

   c). Hemodilution d). Polycythemia

2. Damage to the cilia of the trachea will primarily affect: [K] / [CO2]

   a). Gas exchange b). Voice production

   c). Removal of dust particles d). Oxygen transport by blood

3. Non-volatile acids are produced mainly from metabolism of: [U] / [CO3]

   a). Carbohydrates b). Lipids

   c). Proteins d). Vitamins

4. The second stage of erythropoiesis in fetal life [K] / [CO1]

a). Mesoblastic stage b). Hepatic stage

c). Myeloid stage d). Follicular stage

5. During acclimatization Hb content [U] / [CO2]

a). Falls b). Lowers

c). Decreases d). Increases

6. Bile is produced by the: [U] / [CO3]

a). Gall bladder b). Pancreas

c). Liver d). Small intestine

[1x6 = 6]

Part B

Short Answer

Answer any **2** questions

Each question carries **2** marks

7. Explain the role of chemoreceptors in exercise-induced hyperventilation. [U] / [CO2]

8. Define alkalosis. [K] / [CO3]

9. Recall complete heart block. [K] / [CO1]

10. Define hemolysis. [K] / [CO1]

[2x2 = 4]

Part C

Short Essay

Answer any **4** questions

Each question carries **5** marks

11. Discuss about the valves and blood vessels present in human heart. [U] / [CO1]

12. Enumerate and describe the effects of exposure to severe cold. [U] / [CO2]

13. Explain renal failure and its complications. [U] / [CO3]

14. Describe the effects of exercise on blood and blood volume. [U] / [CO1]

15. Explain pneumonia and its effects on respiration. [U] / [CO2]

16. Explain heart failure on the basis of side of heart affected. [U] / [CO1]

[5x4 = 20]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

17. Explain the types of apnea and describe the pathophysiology and effects of sleep apnea [U] / [CO2]

18. Explain the structure of nephron with the help of a labelled diagram. [U] / [CO3]

19. Define hemostasis and enumerate the stages involved in prevention of blood loss. [K] / [CO1]

20. Explain iron distribution, absorption, transport and storage in our body. [U] / [CO1]

[10x2 = 20]